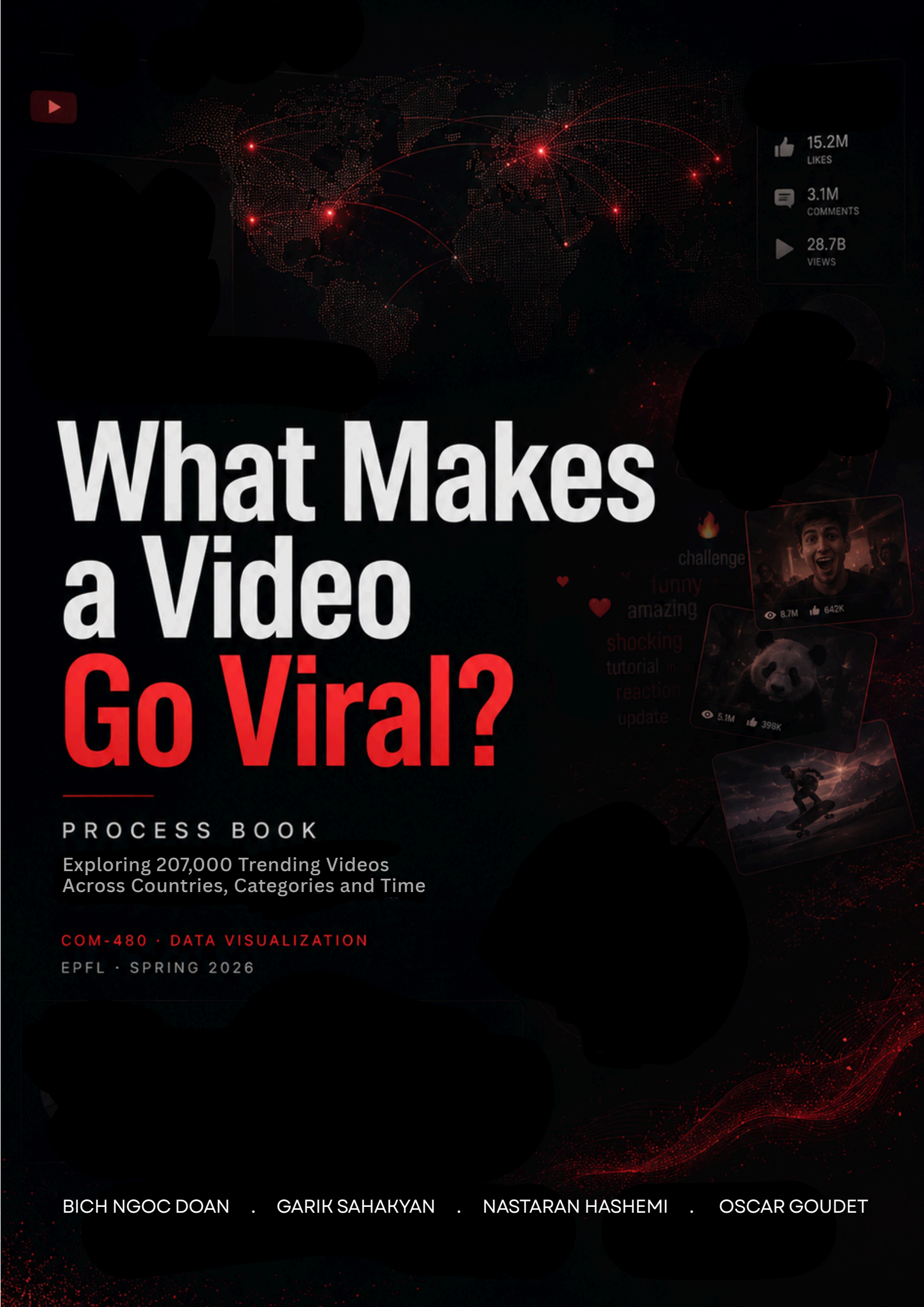




What Makes a Video Go Viral?

PROCESS BOOK

Exploring 207,000 Trending Videos
Across Countries, Categories and Time

COM-480 · DATA VISUALIZATION

EPFL · SPRING 2026

15.2M
LIKES

3.1M
COMMENTS

28.7B
VIEWS

challenge
funny
amazing
shocking
tutorial
reaction
update



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01. INTRODUCTION

DECODING THE PATTERNS OF VIRALITY

Every minute, hundreds of hours of video content are uploaded to YouTube, yet only a tiny fraction appears on the platform's trending page. While the trending tab presents the outcome of virality, the mechanisms behind it often remain opaque. Why do certain videos spread rapidly while others disappear almost instantly? Are viral patterns universal, or do they differ across countries, cultures, and content categories?

These questions formed the starting point of our project.

Using the *Trending YouTube Video Statistics dataset*, we set out to explore how virality manifests across geography, engagement, content, and time. Rather than treating virality as a random phenomenon, we aimed to reveal the recurring structures hidden beneath trending behavior: differences in regional content preferences, engagement dynamics, publishing habits, and temporal patterns.

We approached the project through a dual-audience philosophy:

- **For Content Creators:** the platform acts as an analytical sandbox for exploring posting patterns, engagement benchmarks, and content characteristics across regions.
- **For General Users:** the website provides a more playful and narrative-driven experience that demystifies how digital attention and viral culture are shaped online.

Rather than designing a static dashboard, we aimed to create an interactive storytelling experience driven by exploration and participation. Users progressively move from global overviews to detailed video-level insights by navigating across countries, comparing categories, investigating engagement patterns, and eventually test their own intuition about virality through interactive game-like components. Inspired by Ben Shneiderman's Visual Information-Seeking Mantra: "*overview first, zoom and filter, then details-on-demand*", the interface was designed to encourage active discovery rather than passive consumption of information.

Our central narrative gradually evolved into the idea of a "**recipe of virality**." While no single formula guarantees online success, the dataset reveals recurring tendencies and behavioral patterns that emerge repeatedly across contexts. Through interactive visualization, our goal was therefore to make these hidden dynamics visible, explorable, and engaging to understand.



02. DATASET & CONCEPT DEVELOPMENT

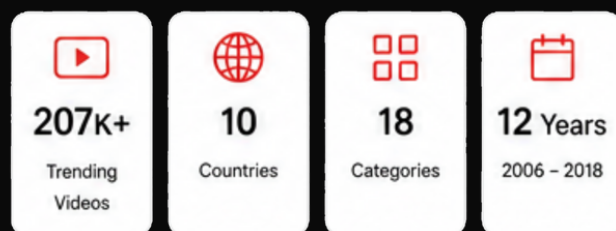
FROM RAW DATA TO VISUAL INSIGHTS

Behind every trending video lies a network of temporal, geographic, and engagement-driven patterns. Before building the interactive experience, we first transformed global YouTube data into a structured system capable of revealing the hidden dynamics of virality.

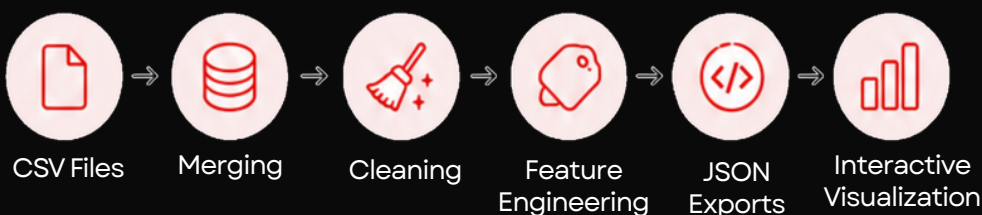
02. 01 DATASET & PREPROCESSING

Undrestanding the Dataset

The project is based on the Trending YouTube Video Statistics dataset, which contains daily snapshots of videos appearing on YouTube's trending page across multiple countries between 2006 and 2018. The dataset includes engagement metrics, publication metadata, and category information, making it particularly suitable for exploring how virality differs across geography, content type, and time.



DATASET PREPROCESSING PIPELINE



Before visualization, all regional CSV files were merged into a unified dataset. Preprocessing involved resolving encoding inconsistencies in several regional files, converting non-standard date formats, And handling missing metadata,. Additional temporal features, such as publication hour, weekday, and days-to-trending, were also engineered to support the interactive analyses and visualizations developed throughout the project.

TECK STACK



Python
Preprocessing
& analysis



D3.js
Interactive
visualizations



JavaScript
Interaction
logic



HTML5 & CSS3
Layout & styling

02. 02 FROM CONCEPTS TO FEATURES

Defining the Viral Recipe

After structuring the dataset, we translated our initial questions into a set of interactive features. Rather than presenting the data as isolated charts, we organized the website around a progressive exploration of virality: first at the global level, then through country/category comparisons, and finally through playful user interaction.

FINAL FEATURES LIST

1. Global Stage

A multi-view geographic exploration in form of an interactive world map introducing country-level differences in trending behavior. Users can compare regions through metrics such as average views, video volume, engagement, and publication time using filters.

The map is complemented by a linked category heatmap matrix, allowing users to compare the relative distribution of trending content categories across countries simultaneously. Together, these coordinated visualizations provide both a geographic overview and a category-level comparative perspective of global virality patterns.

Goal: Provide an overview of how virality varies geographically and establish a high-level perspective before inviting users to move toward more detailed exploration.

2. Explore Section: Choose Your Lens & The Viral Formula

2.1. Choose Your Lens

A more detailed country/category exploration interface where users can select a country and content category to inspect engagement metrics, timing patterns, and representative examples. Interactive filtering allows users to progressively narrow their focus and compare how different niches behave across regions.

Goal: move from global patterns to more specific “viral recipes” for different audiences.

2.2. The Viral Formula

A narrative-driven exploration space bringing together title patterns, rhetorical hooks, engagement metrics, publication timing, thumbnail examples, and audience behavior to help users understand recurring ingredients of trending videos across countries and niches. This section integrates multiple linked visualizations, including word clouds, timing heatmaps, summary metrics, thumbnail carousels, and the interactive “Winning Hook” system. Within the “Winning Hook” visualization, users can explore dominant strategies through proportional distributions and interactively inspect real trending title examples associated with each hook category. Color, ranking, and progressive filtering are used throughout the interface to support comparison and interpretation of viral patterns.

Goal: transform raw engagement data into understandable patterns of viral behavior.

3. Viral Duel

An interactive game experience where users compare two trending videos and predict which one achieved higher view counts. The system provides immediate feedback, and streak tracking to encourage participation and reinforce discovered virality patterns.

Goal: transform passive analysis into active participation and reinforce insights through play.



03 . DESIGN ITERATIONS & CHALLENGES

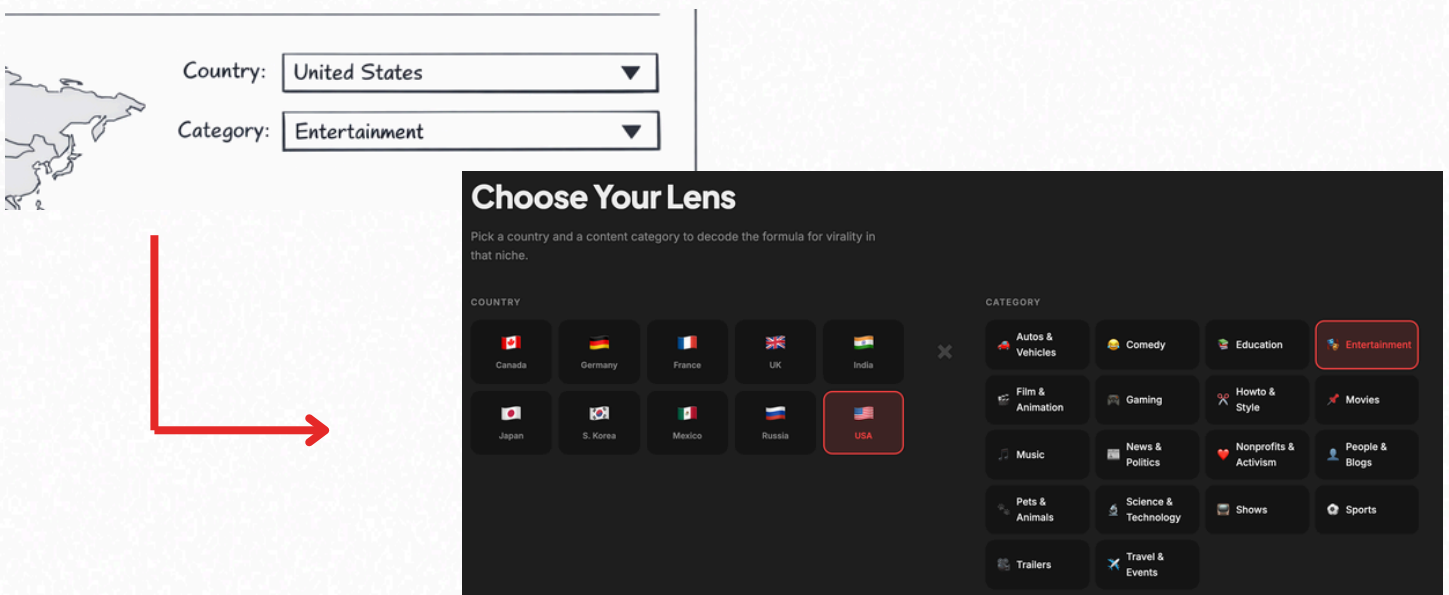
FROM COMPLEX IDEAS TO CLEAR INTERACTIONS

The project initially began as a broad attempt to decode virality across multiple dimensions, combining geographic comparisons, engagement patterns, rhetorical analysis, thumbnail exploration, and interactive simulations. As development progressed, one of the main challenges became transforming this large collection of ideas into a coherent storytelling experience rather than a fragmented collection of charts.

To address this, the interface was designed as a guided progression where users gradually “unlock” deeper patterns of virality through interaction, filtering, and exploration, transforming passive observation into active discovery.

03 . 01 SIMPLIFYING THE INTERFACE

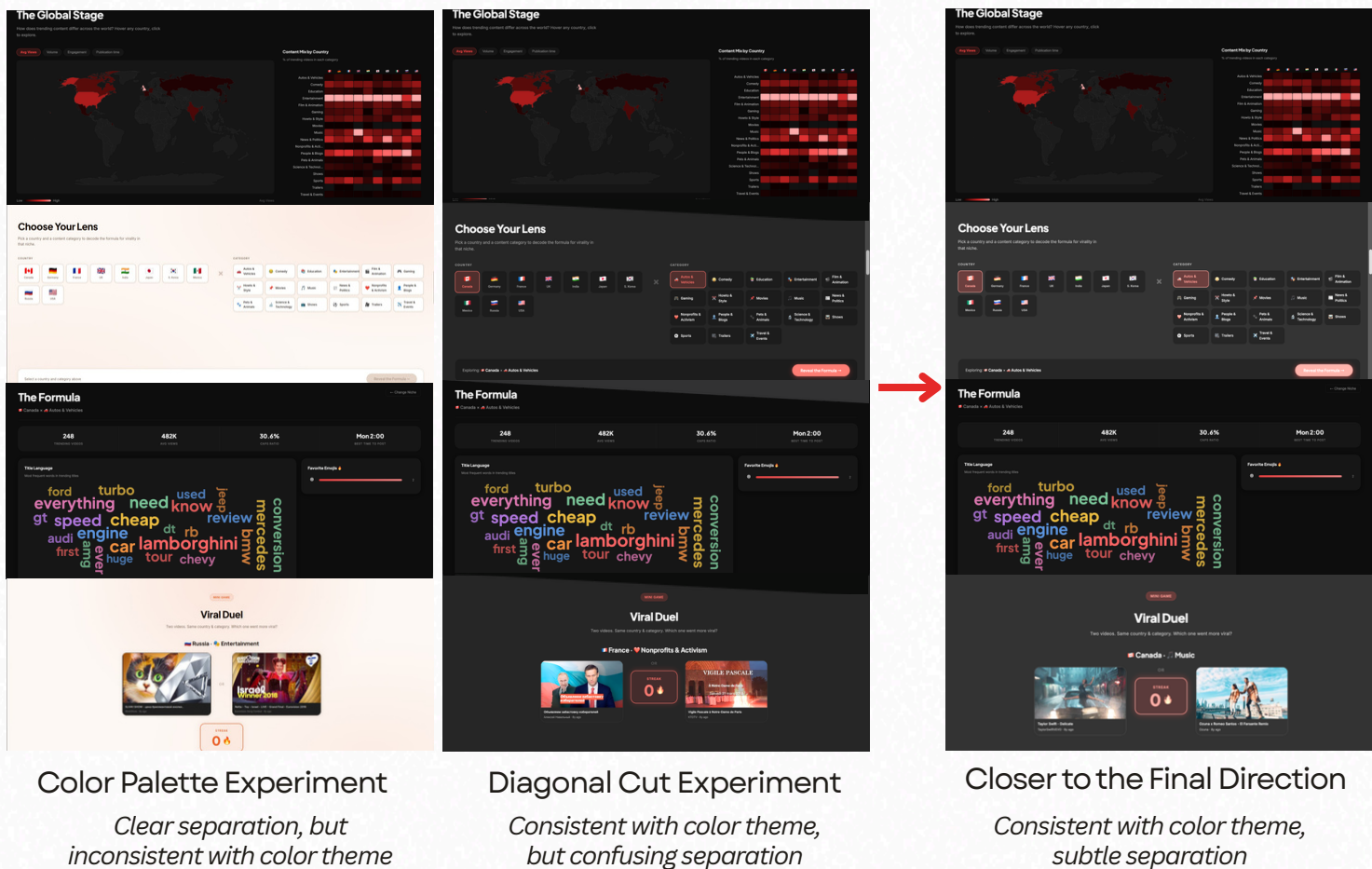
One of the earliest challenges involved navigation and visual complexity. Initially, the project relied heavily on dropdown systems directly integrated into the map visualization for selecting countries and content categories. However, as the number of available combinations increased, this interaction model quickly became visually overwhelming and difficult to navigate. To address this issue, we introduced the dedicated “Choose Your Lens” section, allowing users to transition more naturally from global exploration toward niche-specific analysis. This separation significantly improved readability, reduced interface clutter, and reinforced the storytelling structure of the website.



03.02 STRUCTURING THE USER JOURNEY

A central challenge was maintaining a clear sense of narrative progression as users moved through the experience. Early versions used a uniform black background throughout the interface, causing sections such as the "The Global Stage", "Choose Your Lens," and "The Viral Formula" to visually blend together and made it difficult for users to distinguish where one narrative segment ended and another began.

To address this, we iterated through three solutions. First, we introduced background color separation between sections. Two approaches were tested: alternating **Color Palette** (black-lightOrange-black) among sections (clear separation, but inconsistent with the project's aesthetic identity) and a **Diagonal Cut** between black and dark grey (more on-brand, but too subtle to be reliably legible). We settled on a gentler black-to-grey distinction, but the narrative flow still lacked the level of segmentation and progression we wanted.



So we developed the third and final alteration, focused on progressive narrative revelation. Rather than exposing all visualizations simultaneously, users are sequentially guided from one section to the next through call-to-action button – "Choose Your Lens", "Reveal the Formula", "Play the Viral Duel" – serves as a deliberate narrative gate that both advances the story and unlocks the corresponding label in the navigation bar.

Now, users always know exactly how far into the experience they have travelled, and can freely revisit any section they have already unlocked.

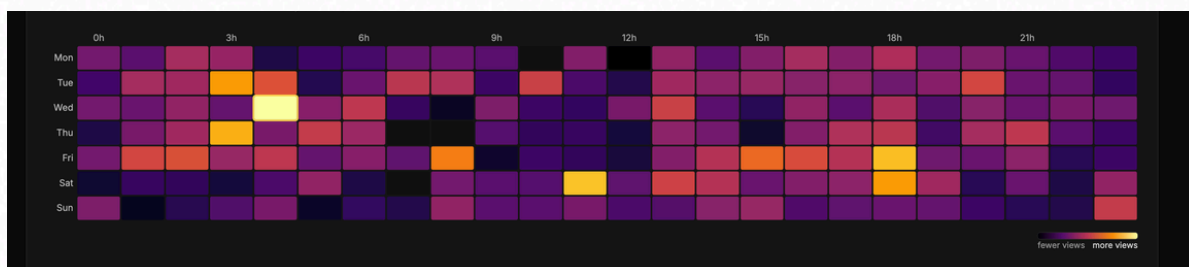


Once all sections unlocked: free navigation anywhere
Try another country x category

03.03 IMPROVING HEATMAP & MAP VISIBILITY

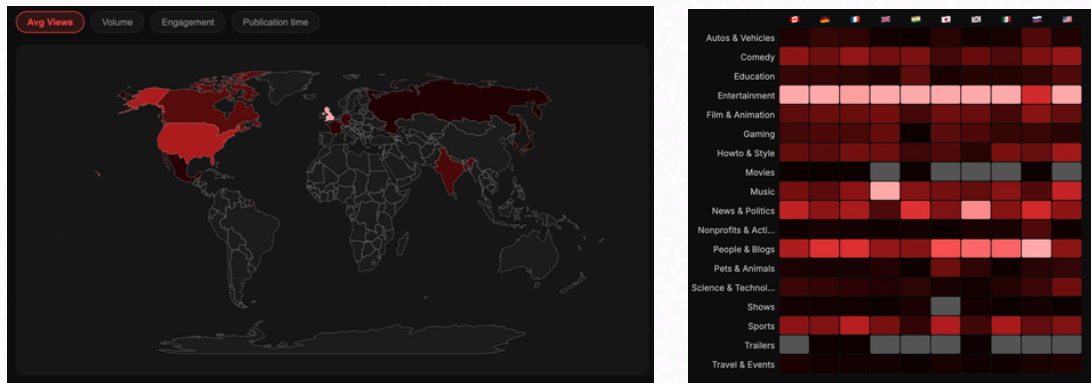
One of the other challenges involved improving the readability of the world maps and heatmaps while preserving the site's black-red visual identity. Early versions relied primarily on red scales over dark backgrounds, making differences difficult to distinguish, particularly in lower ranges.

To improve perceptual separation, we initially experimented with more divergent multi-color palettes. Although these alternatives increased distinction between values, they visually conflicted with the overall aesthetic of the website and introduced unnecessary visual distraction across sections.



Palette Experiment
Higher distinction, but visually distracting

The final solution focused instead on extending the original palette through the introduction of lighter reds, whites, and neutral gray tones. This approach significantly improved contrast and value differentiation while maintaining stylistic consistency with the YouTube-inspired interface. In the geographic visualizations, readability was further improved by thickening country borders and increasing boundary contrast, allowing users to distinguish regions more clearly without sacrificing the dark cinematic atmosphere of the platform.



Final Refined Scale

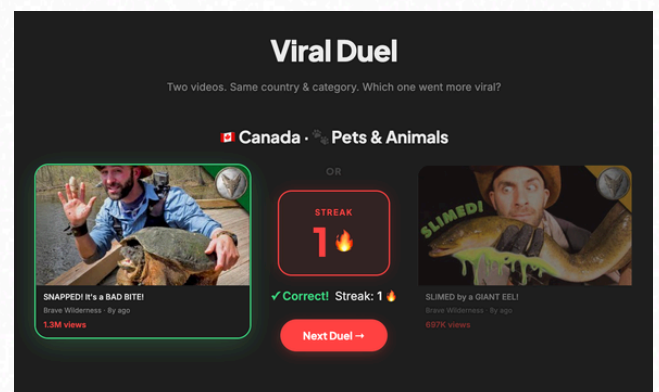
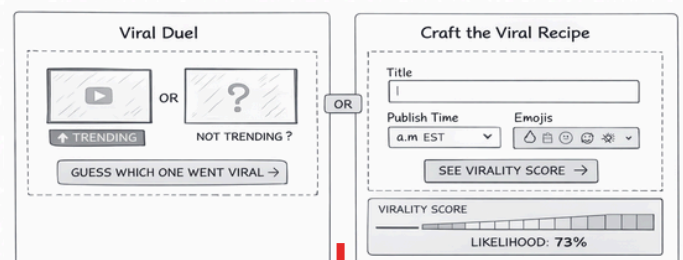
Balanced readability and aesthetic consistency

03.04 ACTIVE LEARNING THROUGH PLAY

As the project evolved, we became interested in transforming users from passive observers into active participants. Rather than ending the experience with analytical dashboards alone, we explored how a game could reinforce users' understanding of virality patterns.

Early concepts explored two different directions. The first focused on a predictive “Craft the Viral Recipe” simulator, where users would manually construct a potentially viral video by selecting elements such as title style, emojis, and publication timing in exchange for a computed “virality score.” Early sketches revealed that the interaction became overly form-drive.

The final solution instead focused on Viral Duel: a lightweight comparison game where users predict which of two real trending videos performed better within the same country and category. By reusing the rhetorical hooks, thumbnails, timing patterns, and engagement cues introduced throughout “The Formula,” the game transforms exploration into a form of active learning and intuitive visual reasoning.



Final Choice – Viral Duel

Focused on rapid comparison and intuitive interaction



04. PEER ASSESSMENT

TEAM WORKFLOW

The development of Chartistry was executed through a collaborative, iterative workflow. All team members participated actively in the major stages of the project, including brainstorming visualization concepts, discussing storytelling approaches, refining interactions, and evaluating design decisions. Most implementation and coordination decisions emerged through team meetings, both in person on campus and remotely through Google Meet sessions. While the project evolved collectively, team members gradually focused on specific components depending on their interests.

The distribution of responsibilities throughout the project can be summarized as follows:

Bich Ngoc Doan

- Contributed to preprocessing of the YouTube trending dataset
- Participated in developing the structure and core skeleton of the website
- Implemented text- and emoji-based analyses and visualizations
- Contributed to debugging, interface refinement, and iterative design discussions

Garik Sahakyan

- Worked on preprocessing and analytical exploration of temporal patterns
- Developed engagement-related analyses and comparative visualizations
- Contributed to refining interactions and improving visualization readability
- Participated in debugging and iterative improvements of the platform

Nastaran Hashemisanjani

- Contributed to preprocessing and conceptual dev of the project narrative
- Designed the sketches and visualization concepts in Miestone 2 report
- Participated in defining the interaction flow and structured the final process book
- Contributed to debugging, interface refinement, and iterative design discussions

Oscar Goudet

- Contributed to preprocessing and implementation of global-level visualizations
- Developed interactive geographic exploration components
- Implementating the virality game section
- Contributed to debugging, interface refinement, and iterative design discussions

